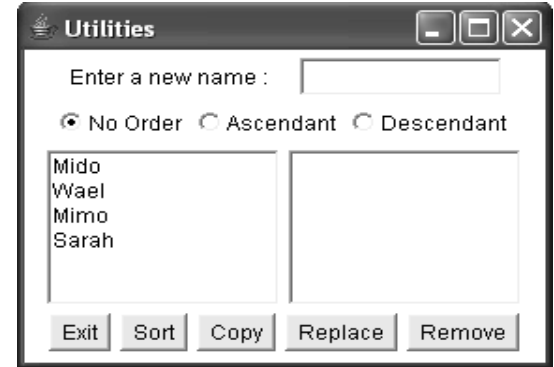


A- Treat ONLY one of the following two questions:

1. Write a program that displays the following window:

Initially, the first list (on the left) contains the following 4 elements as shown in the figure below.

- **Exit** : to exit from the program
- **Sort**: allows sorting the first list (on the left) in ascending or descending order according to the selected option.
- **Copy**: to copy all the elements of the first list to the second list. (on the right).
- **Replace**: allows replacing the selected name of the first list with the new entered name. In case of no selection of any name, the new name is added at the end of the first list.
- **Remove** : to delete the selected name in the first list



(8 pts)

2. Write a Java program that:

- Defines the class "Product" with the attributes: Product name, available quantity, unit price, product category
- Displays the following page:

Product Name :

Available Quantity: Unit price:

Category List:

Electromenager
Meubles
Plantes

- "Create" button allows creating an instance of the class "Product" based on the values entered and selected by the user.
- "Save" button allows saving all created instances in the file "Product.dat"
- "Display" button allows displaying the information of a saved instance in the file "Product.dat" based on the product Name.
- "List" button allows listing the names of products belonging to a specific category.

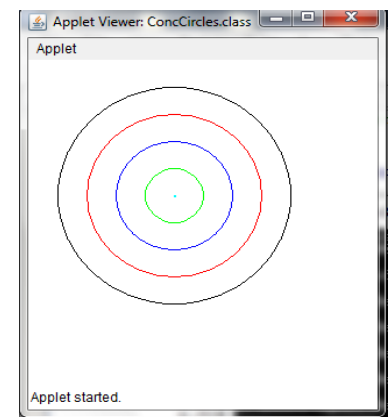
(8 pt)

B- Required Part: (Treat all the following questions)

1. Write a Java Applet that creates a Simple **Timer**. This applet also implements "Runnable". The time is updated every second. When the user displays another page in the browser, the Applet has to be stopped and the seconds, minutes and hours have to be reset on the next appearance of the applet. (5 pts)



2. Write a Java Applet that draws the following figure: each circle has a different color than the other. The biggest circle has a diameter 200 pixels, the second is 150 pixels, the third is 100 pixels, and the smallest is 50 pixels. The distance between each circle is 25 pixels. The height of the Applet is 300 pixels, and the width is 300 pixels. (3 pts)



3. Define a class that inherits from **Thread** and receives a name and an integer (n) as parameters. Its body is a loop of 20 times, displaying its name on the standard output, and then sleeps for n seconds. Write a program that receives as command line argument two integers n1 and n2. This program will create two threads of the previously created class **Thread**, passing n1 to the first and n2 to the second, and then launches the two threads. (4 pts)